

Evaluating Estimators

For a particular estimation problem there may be several (maybe even many) competing estimators. We'll spend the next bit of time answering the questions:

- Are some estimators better, in some sense, than others?
- If so, in what sense?

Right off, we acknowledge that some estimators are indeed better than others. For example, if $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$ we have two competing estimators for λ : $\bar{X}$ and S^2 (why?). It turns out that $\bar{X}$ is a better estimator than is S^2 . To see why, we'll start by discussing some properties of estimators that will help us decide which are better.

Criteria for Good Estimators

💡 “Good” estimators

A good estimator is **unbiased**. An estimator $\hat{\theta}$ is an unbiased estimator of θ if:

$$E(\hat{\theta}) = \theta.$$

In addition to being unbiased, a good estimator has small variance. If $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased estimators of θ , then $\hat{\theta}_1$ is a better estimator (or, a more efficient estimator) than $\hat{\theta}_2$ if:

$$\text{Var}(\hat{\theta}_1) < \text{Var}(\hat{\theta}_2).$$

Remember that estimators are statistics, and are themselves RVs. As such, they have their own distributions.

Bullseye!

If an estimator of a parameter (or function of parameters) is unbiased and has the smallest variance of all possible unbiased estimators of that parameter (or function of parameters), then the estimator is the **best unbiased (or most efficient) estimator** for that parameter (or function of parameters).